

Pick and Protect Fragile Glassware SOP

One episode empties the pick row into one tote. A paper base sheet goes down first, the glasses go in heaviest to lightest with a paper pad standing between every pair, a void pad fills every gap left around them, and the fragile flag goes on the tote last. The episode ends when the pick row is empty and the tote is flagged.

The order never changes: base sheet, the heaviest glass, then a pad and the next heaviest glass again and again until the pick row is empty, then the void pads, then the fragile flag. No glass goes in before the base sheet is flat on the tote floor. No glass goes in before the pad that separates it from the glass already in is standing. No void pad goes in until the pick row is empty, and the fragile flag does not go on until every void is filled.

The gripper on the pick row's side works the pick row and every glass. The right gripper works the right paper tray and the fragile flag. The left gripper works the interleave pads and the left paper tray. The base sheet goes to the gripper on its side. Both grippers reach into the tote at front center, each from its own side. Neither gripper takes the other's zones.

Glass breaks when it is squeezed, dropped, or knocked, so every glass is held only hard enough to stop it moving, carried upright and low over the desk, and lowered until it is resting before the gripper opens. A glass never touches another glass, never touches a tote wall, and is never pressed, tapped, nudged, or leaned on. The tote, the pick row, and the two paper trays stay where they are put.

The desk is set up in one of three ways. Only the pick row moves; the tote, the pad row, the two paper trays, and the flag spot are in the same place in all three. The base sheet moves only when the pick row takes its place.

- **Config L:** the pick row is at the back left; the base sheet lies at the back right instead.
- **Config M:** the pick row is at the back center, behind the tote.
- **Config R:** the pick row is at the back right.

Where a step depends on the setup it says so on an **IF** line — look at the desk and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the pick row starts at the back left (**Config L**), the back center (**Config M**), or the back right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the pick row's side takes every glass — the left gripper in Config L and M, the right gripper in Config R. The gripper on the base sheet's side lays it. No arm reaches across the desk.

- **Fixed roles:** the left gripper takes the interleave pads and the left void pads, the right gripper takes the right void pads and the fragile flag, in every config.
- **Order:** base sheet, heaviest glass, then pad and next glass until the pick row is empty, then the void pads, then the flag, in every config.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole desk: the pick row in the start zone for this episode's config, the flag spot at the middle of the right edge, the right paper tray at the front right, the base sheet at the back left (back right in Config L), the interleave pads at the middle left, the left paper tray at the front left, and the tote at front center.
3. The whole pick row is in frame, so it can be seen going empty, and the tote interior is visible from above, so every glass can be seen going in.
4. Both arms are at home with grippers open.
5. The desk is bare apart from the glasses, the tote, the base sheet, the interleave pads, the two paper trays, and the fragile flag. There are no marks, mats, or fixtures on it.
6. Nothing obstructs the paths between the pick row and the tote. The right arm reaches (Config R) every glass in the pick row, (Config L) the base sheet, the flag spot, the right paper tray, and the tote's right side and back wall. The left arm reaches (Config L and M) every glass in the pick row, (Config M and R) the base sheet, every interleave pad, the left paper tray, and the tote's left side and back wall.

Materials checklist

1. Glasses stand upright in one row across the back of the desk, the **pick row**, each on its own base with a clear gap to its neighbors, so a gripper closes on one without touching another. They are clean, dry, and free of chips and cracks. The row stands in the start zone for this episode's config, and the other two start zones are bare:
 - **Config L:** back left, where the base sheet normally lies
 - **Config M:** back center, behind the tote, clear of the base sheet and the flag spot
 - **Config R:** back right

2. The pick row holds glasses of visibly different weights, so **heavy glasses**, **medium glasses**, and **light glasses** are told apart at a glance by shape. Glasses of the same weight are identical to each other.
3. The pick row holds no more glasses than fit in the tote in one row from the back wall toward the front, standing upright with a pad between every pair and none touching.
4. One open top **tote** at front center, square to the front edge, empty and dry. Its walls are low enough that a glass is lowered in from above without the gripper touching the rim. It has a flat **label ledge** along the top of its near wall.
5. One **base sheet** of kraft paper, cut to the tote floor, lying flat at the back left, or at the back right in Config L. It is stiff enough to keep its shape when it is pinched at one edge and lifted.
6. **Interleave pads** lie in a row at the middle left, more of them than there are gaps to fill between the glasses in the pick row. Each is a kraft sheet folded into a pad with a flat foot, so it stands on its own on a flat surface without support. Each pad is as tall as the tallest glass and as wide as the tote floor.
7. Two shallow **paper trays**, the **left paper tray** at the front left and the **right paper tray** at the front right, each holding a small pile of **void pads**: loosely crumpled paper balls, each wider than the gaps it fills and soft enough to squash down as it goes in.
8. One printed **fragile flag**, a stiff card that keeps its shape when pinched at an edge, lying printed side up at the **flag spot** on the middle of the desk's right edge.

Workspace layout

Every zone is a position on the bare desk. Nothing is marked, so each position is held by where the thing was set.

- **Pick row**: one row of glasses standing upright with clear gaps between them, at the back left (Config L), the back center (Config M), or the back right (Config R).
- **Flag spot**: the middle of the desk's right edge, between the back right and the right paper tray.
- **Right paper tray**: the front right, holding void pads.
- **Base sheet spot**: the back left; the back right in Config L.
- **Pad row**: the middle left, the interleave pads in a row.
- **Left paper tray**: the front left, holding void pads.
- **Tote**: front center, square to the front edge, in reach of both arms. It stays on the desk for the whole episode.

Each thing goes into the tote from the side it is picked on: the interleave pads and the void pads from their own side, the glasses from the pick row's side (straight forward in Config M), the flag from the right. No glass crosses the middle of the desk and no glass passes over another glass.

Vocabulary

- **Arm assignment:** the right arm takes the void pads in the right paper tray and the fragile flag, and in Config R every glass off the pick row and in Config L the base sheet. The left arm takes every interleave pad and the void pads in the left paper tray, and in Config L and M every glass off the pick row and in Config M and R the base sheet. Neither arm takes the other's zones.
- **Pick row:** where the glasses stand at the start of the episode — the back left (**Config L**), the back center (**Config M**), or the back right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Supply side:** the side of the desk holding the base sheet — the left in Config M and R, the right in Config L. The gripper on that side lays it.
- **Heavy glass:** a thick walled glass, short and wide on a heavy base. Heavy glasses go in first.
- **Medium glass:** a straight sided glass, thinner walled and taller than a heavy glass. Medium glasses go in after the heavy ones.
- **Light glass:** a thin walled glass, the tallest and narrowest kind. Light glasses go in last.
- **Weight order:** heaviest at the tote's back wall, lightest at the front. Each time, the heaviest glass still standing in the pick row is the next one in, so the light glasses end up at the front where nothing leans on them. When two glasses in the pick row weigh the same, take the one nearer the tote first.
- **Base sheet:** the flat kraft sheet that covers the tote floor. It goes in before any glass and every glass stands on it.
- **Interleave pad:** the folded kraft pad that stands upright on the base sheet between two glasses. It goes in before the glass that comes after it, and it keeps glass from touching glass.
- **Standing:** the interleave pad rests on its own foot, upright, spanning the tote from side wall to side wall, and stays that way with no gripper on it.
- **Void:** a gap between a glass and a tote wall that is wide enough to see the tote floor through. A void runs along a side wall, or across the front of the tote when the last glass in stops short of the front wall.
- **Void pad:** a crumpled paper ball that fills a void.
- **Filled:** the void pad sits down in the gap, level with or below the glass rims, with no tote floor visible in that gap, and no glass moved by it going in.
- **Glass grasp:** the gripper closes on the glass body low down, near its base, taking the wall on two opposite sides, with the glass upright. Never on the rim, never over the mouth, never on the base underneath.
- **Soft close:** close the gripper until the glass stops moving in it, then stop closing. No squeeze after that, and no tightening once the glass is off the desk. If the glass slips, lower it

back down onto a clear spot and take it again.

- **Lift:** taking a glass or a piece of paper straight down onto its grasp, closing, and raising it straight up until there is daylight under it, before anything moves sideways.
- **Low carry:** carrying the glass upright and level, just clear of the desk and just clear of the tote rim, so it is never high above anything it could land on.
- **Lower in:** carrying the thing level over the tote, lowering it straight down until it is resting, and releasing once it is resting. Nothing is ever dropped in from above the rim.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing rocking, sliding, tipping, or leaning.
- **Upright:** the glass stands on its own base on the base sheet, mouth up, not leaning on a pad, a wall, or another glass.
- **Pick row empty:** no glass is left standing on the desk in the pick row's start zone.
- **Release point:** over the spot in the tote, or over the label ledge, that the thing in the gripper is going to. A gripper opens nowhere else.
- **Label ledge:** the flat top of the tote's near wall, where the fragile flag rests printed side up, square to the front edge and clear of the tote's opening.

Steps

Steps 1 to 6 are one episode and one tote. Every glass is lifted straight up off the pick row and carried upright in the orientation it will be set down in.

Step 1 and the glass picks in Steps 2 and 3.2 depend on the config: in Config L and M the **left gripper** takes each glass from the pick row, in Config R the **right gripper** does; in Config L the base sheet lies at the back right and the **right gripper** lays it. Every other line is the same in all three configs.

Step 1: Line the tote

Goal: the base sheet lying flat on the tote floor, corner to corner.

- **IF Config M or R:** the **left gripper** pinches the near edge of the **base sheet** at the back left.
IF Config L: the **right gripper** pinches it at the back right.
- Lift it straight up, carry it level to the **tote**, lower it flat onto the tote floor, and release once it is resting.
- Lay the sheet in one piece. Never fold it, never tear it, and never push it down a tote wall to flatten it.

Check: the gripper is clear and the base sheet lies flat on the tote floor, reaching the walls on all four sides, with no fold, bunch, or curled corner. If it lands bunched or short of a wall, lift it clear by its near edge and lay it again.

Step 2: Nest the first glass

Goal: the heaviest glass in the pick row standing upright on the base sheet against the tote's back wall.

- Look along the **pick row** and find the heaviest glass standing in it.
- **IF the pick row is at the back left (Config L):** the **left gripper** comes down on that glass, takes the **glass grasp** low on its body, uses a **soft close**, and carries it to the tote's left side.
- **IF the pick row is at the back center (Config M):** the **left gripper** comes down on that glass, takes the **glass grasp** low on its body, uses a **soft close**, and carries it straight forward to the tote.
- **IF the pick row is at the back right (Config R):** the **right gripper** comes down on that glass, takes the **glass grasp** low on its body, uses a **soft close**, and carries it to the tote's right side.

Then, in all three:

- **Lift** it straight up until there is daylight under it, then **low carry** it upright and level to the **tote**.
- **Lower in** against the tote's **back wall**, until the glass is resting on the base sheet, and release.

Take it whichever spot it stands in on the pick row. Never move another glass aside to reach it.

Check: the gripper is clear and the glass stands **upright** and **settled** on the base sheet against the back wall, not leaning on the wall and not rocking. If it leans or rocks, take the **glass grasp** again, lift it straight up, and lower it in again.

Step 3: Interleave and nest the rest of the pick row

Goal: the **pick row empty** and every glass standing upright in the tote in **weight order**, each pair separated by a standing interleave pad.

If the **pick row** is empty, go to Step 4. If a glass is still standing in it, run 3.1 and 3.2, then read this line again. Finish both substeps for one glass before starting the next glass.

3.1 Stand the interleave pad

- The **left gripper** pinches the top edge of the next **interleave pad** in the pad row, lifts it straight up, and carries it level to the **tote**.
- **Lower in** just in front of the glass that went in last, until the pad's foot is resting on the base sheet, and release once it is **standing**.
- The pad goes down on its own foot with a clear gap to the glass behind it. Never lean it on that glass and never press it against it.

Check: the **left gripper** is clear and the pad is **standing** across the tote from side wall to side wall, upright, touching no glass. If it falls or leans, pinch its top edge, lift it clear, and stand it again.

3.2 Nest the next glass

- Look along the **pick row** and find the heaviest glass still standing in it. When two weigh the same, take the one nearer the tote.
- **IF Config L or M:** the **left gripper** comes down on that glass. **IF Config R:** the **right gripper** does. Take the **glass grasp** low on its body and use a **soft close**.
- **Lift** it straight up, **low carry** it upright and level to the **tote**, and **lower in** just in front of the standing pad until it is resting on the base sheet.
- Release once it is resting. Lower it into the open space in front of the pad, not against the pad and not onto it.

Check: the gripper is clear and the glass stands **upright** and **settled** on the base sheet, with the pad between it and the glass behind it, and no glass touching another glass. If the glass leans, or the pad has been pushed over, lift the glass clear, stand the pad again, and lower the glass in again.

Expected state after Step 3: the pick row is empty. The tote holds, from the back wall forward, the heaviest glass, then a standing pad and the next heaviest, and so on to the lightest glass at the front, all standing upright on the base sheet with a pad between every pair and nothing touching anything else.

Step 4: Pad the voids

Goal: every void filled, with no glass moved.

- The **left gripper** pinches a **void pad** from the **left paper tray**, lifts it straight up, carries it level to the **tote**, and lowers it into the gap along the tote's **left side wall** until it is **filled**, then releases.
- The **right gripper** pinches a **void pad** from the **right paper tray**, lifts it straight up, carries it level to the **tote**, and lowers it into the gap along the tote's **right side wall** until it is **filled**, then releases.
- If tote floor still shows across the **front** of the tote, the **right gripper** takes another void pad from the right paper tray and fills that gap the same way.
- If tote floor still shows anywhere along a side wall, the gripper on that side takes another void pad from its own tray and feeds it down until that gap is **filled** too.
- Feed each pad down into its gap and let it squash as it goes. Never push a pad against a glass to make it fit, and never pack it down on top of the glasses.

Check: both grippers are clear of the tote and the tote is in frame. No tote floor is visible anywhere around the glasses, every glass still stands **upright**, and no pad rides above the

glass rims. If a pad has moved a glass, lift the pad out, stand the glass again, and feed the pad back down.

Step 5: Flag the tote fragile

Goal: the fragile flag lying printed side up on the tote's label ledge.

- The **right gripper** pinches the near edge of the **fragile flag** at the flag spot, lifts it straight up, and carries it level to the **tote**.
- Lower it flat onto the **label ledge**, printed side up and square to the front edge, and release once it is resting.

The flag goes on only after every void is filled. Never lay it inside the tote, on a glass, or on a void pad.

Check: the **right gripper** is clear and the flag lies flat and printed side up on the label ledge, readable in frame, square to the front edge, clear of the tote's opening, and resting on nothing but the ledge. If it hangs off the ledge or lands printed side down, pinch its near edge, lift it clear, and lay it again.

Step 6: Confirm the pack and end the episode

- Confirm the tote holds the base sheet on its floor, every glass standing upright in weight order from the back wall forward, a standing pad between every pair, every void filled, and the fragile flag printed side up on the label ledge.
- Confirm the **pick row empty** and the desk otherwise bare.
- Return both arms home, clear of the tote and the pick row, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the fragile flag off the label ledge and lay it printed side up at the flag spot.
2. Take the void pads out of the tote. Uncrumple and recrumple each one and drop it back in a paper tray, splitting them between the left and the right tray. Replace any pad that has gone limp or torn, and top both trays back up so each holds a small pile.
3. Lift the glasses out one at a time, front to back, and set them upright in the pick row for the next episode's config — back left (Config L), back center (Config M), or back right (Config R) — with a clear gap between neighbors.
4. Take the interleave pads out and lay them back in the pad row at the middle left, with the pads that were not used. Refold any pad whose foot no longer holds it standing on its own, and replace one that is creased soft, torn, or damp. Confirm the pad row holds more pads than there are gaps to fill between the glasses now in the pick row.

5. Lift the base sheet out and lay it flat at the back left, or at the back right if the next episode is Config L. Replace it if it is torn, bent, or will not lie flat.
6. Set the empty tote back at front center, square to the front edge, and wipe out any paper scraps.
7. Vary which glass stands where in the pick row between episodes rather than rebuilding the same row every time, and vary the mix of weights in it. The pick order stays heaviest first and lightest last.
8. Confirm the pick row still holds no more glasses than fit in the tote in one row from the back wall toward the front, standing upright with a pad between every pair and none touching.
9. Check every glass in the light: replace any glass that is chipped, cracked, starred, or will no longer stand on its own base. A chipped glass never goes back in the pick row.
10. Wipe the desk and confirm it is dry and bare. A slick, wet, or crumbed desk lets a glass and the tote slide.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R (pick row at the back right, base sheet at the back left). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the desk — the pick row is not in the start zone for the config; the base sheet is not on the side that config puts it; a gripper reaches across the desk for a glass or the base sheet; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the gripper on the pick row's side takes every glass, the gripper on the base sheet's side lays it; no arm reaches across

the desk; the IF line followed is the one for the config on the desk).

- **Coaching note:** look where the pick row and the base sheet are before the first reach, then follow that config's IF lines through Steps 1, 2, and 3.2.

Violation: Wrong order

- **Visible cue:** a glass goes in before the base sheet is flat on the tote floor, a glass goes in before its interleave pad is standing, a void pad goes in while a glass is still standing in the pick row, or the fragile flag goes on while tote floor is still visible around the glasses.
- **SOP rule broken:** Steps 1 to 5, the order never changes.
- **Coaching note:** finish the step you are in before you start the next one.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes a glass, the fragile flag, or a void pad from the right paper tray; or the right gripper takes the base sheet, an interleave pad, or a void pad from the left paper tray.
- **SOP rule broken:** Steps 1 to 5, the right arm takes the glasses, the right void pads, and the flag; the left arm takes the base sheet, the interleave pads, and the left void pads.
- **Coaching note:** paper from the left, glass from the right. Nothing crosses the middle.

Violation: Glasses nested out of weight order

- **Visible cue:** a glass is taken while a heavier one is still standing in the pick row, or the finished tote reads front to back as anything but lightest to heaviest.
- **SOP rule broken:** Steps 2 and 3.2, the heaviest glass still in the pick row is always the next one in.
- **Coaching note:** heaviest to the back wall first, lightest to the front last, every time.

Violation: Base sheet skipped or badly laid

- **Visible cue:** the first glass goes onto a bare tote floor, or the base sheet lies folded, bunched, curled at a corner, or short of a tote wall when the first glass goes in.
- **SOP rule broken:** Step 1, the base sheet lies flat on the tote floor corner to corner before any glass goes in.
- **Coaching note:** flat and to all four walls, then start packing.

Violation: Interleave pad skipped

- **Visible cue:** a glass is lowered in with no pad standing between it and the glass behind it, or the finished tote has a pair of glasses with no pad between them.
- **SOP rule broken:** Step 3.1, a pad stands between every pair of glasses.
- **Coaching note:** pad first, then the glass. Every time, no exceptions.

Violation: Interleave pad not standing

- **Visible cue:** a pad is laid flat on the base sheet, leaned against a glass, pressed against a glass, dropped in folded, or left fallen while the next glass goes in.
- **SOP rule broken:** Step 3.1, the pad rests on its own foot, upright, wall to wall, touching no glass.
- **Coaching note:** stand it on its foot and let go. If it falls, stand it again before the glass.

Violation: Glass touched a glass, a pad, or a wall

- **Visible cue:** two glasses meet, a glass is lowered against a standing pad or knocks it over, a glass is lowered leaning on the tote wall, or a glass rings against anything in the tote.
- **SOP rule broken:** Steps 2 and 3.2, each glass is lowered into open space and stands upright touching nothing.
- **Coaching note:** find the open space before you go down. Glass never touches glass.

Violation: Glass taken by the wrong part

- **Visible cue:** the gripper closes on a glass's rim, over its mouth, under its base, or high on the body instead of low near the base.
- **SOP rule broken:** Steps 2 and 3.2, take the glass grasp low on the body, on two opposite sides, with the glass upright.
- **Coaching note:** low on the body, every glass. Land on the grasp or reset and take it again.

Violation: Force put on a glass

- **Visible cue:** the gripper keeps closing after the glass has stopped moving, tightens on a glass in mid carry, presses or taps a glass down into the tote, or leans on a glass with a closed gripper.
- **SOP rule broken:** Steps 2, 3.2, and 4, use a soft close and stop, and never press a glass.
- **Coaching note:** hold it just enough to move it. Nothing more, ever.

Violation: Glass carried high or tilted

- **Visible cue:** a glass rides well above the desk or the tote rim in transit, tips out of upright, or is swung rather than carried level.
- **SOP rule broken:** Steps 2 and 3.2, low carry the glass upright and level, just clear of the desk and the tote rim.
- **Coaching note:** low and level. A short fall breaks less than a long one.

Violation: Glass dragged instead of lifted

- **Visible cue:** a glass slides along the desk, the pick row, or the base sheet still touching it, with no daylight under it in transit.
- **SOP rule broken:** Steps 2 and 3.2, lift straight up until there is daylight before moving sideways.

- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Turned or regripped in the air

- **Visible cue:** a glass is turned, tipped, or rolled during the carry, or the gripper shifts, rolls, or seats its hold again on a glass, a pad, or the flag without setting it down first.
- **SOP rule broken:** Steps 1 to 5, carry each thing in the orientation it will be set down in, and set it down before taking a new hold.
- **Coaching note:** orientation is set at the pick. Put it down before you take a new hold.

Violation: Dropped instead of lowered

- **Visible cue:** a gripper opens above the tote rim and the glass, pad, sheet, or flag falls the rest of the way in, or a glass is released before it is resting on the base sheet.
- **SOP rule broken:** Steps 1 to 5, lower in until it is resting, then release.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the desk, over the pick row, over a paper tray, or over open space, and the thing lands anywhere but the spot it was going to.
- **SOP rule broken:** Steps 1 to 5, a gripper opens over the release point and nowhere else.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Glass left leaning or rocking

- **Visible cue:** the episode goes on with a glass tilted, leaning on a pad, a wall, or another glass, or still rocking when the gripper lifts clear.
- **SOP rule broken:** Steps 2 and 3.2, every glass stands upright and settled on the base sheet before the next thing goes in.
- **Coaching note:** watch it settle before you move on.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, taps, or slides a glass, a pad, a void pad, or the flag into position after releasing it, instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 1 to 5, correct a bad placement by lifting the thing clear and lowering it in again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Voids left unfilled

- **Visible cue:** the episode goes on with tote floor visible along a side wall or across the front of the tote, or with a gap left unfilled while void pads are still in the trays.
- **SOP rule broken:** Step 4, every void is filled before the flag goes on.

- **Coaching note:** look all the way round the glasses. If floor shows, feed a pad down.

Violation: Void pad forced in

- **Visible cue:** a pad is pushed against a glass to make it fit, packed down on top of the glass rims, or a glass shifts, tilts, or is knocked over as a pad goes in.
- **SOP rule broken:** Step 4, feed each pad down its gap and let it squash. Never push it against a glass.
- **Coaching note:** the paper gives way, the glass does not. Feed it down the gap.

Violation: Fragile flag mishandled

- **Visible cue:** the flag lands printed side down, inside the tote, on a glass, on a void pad, hanging off the ledge, folded or creased, or is moved again after it is laid down; or the episode ends with the flag still at the flag spot.
- **SOP rule broken:** Step 5, the flag lies flat and printed side up on the label ledge and stays there.
- **Coaching note:** printed side up on the ledge, then leave it alone.

Violation: Two things handled at once

- **Visible cue:** a gripper holds two glasses, or a glass and a pad, at the same time; both grippers close on the same glass; or the left gripper starts the next pad while the right gripper is still carrying a glass.
- **SOP rule broken:** Steps 1 to 5, one thing per gripper, and each substep finishes before the next one starts.
- **Coaching note:** one thing at a time. Finish it, then start the next.

Violation: Check skipped

- **Visible cue:** the next thing goes in with a gripper still over the tote so the placement cannot be seen, or Step 4 ends with no still moment where both grippers are clear and the filled tote is in frame.
- **SOP rule broken:** the checks in Steps 1 to 5, get the gripper clear and confirm the state before moving on.
- **Coaching note:** get clear and look. The check is part of the step.

Violation: Tote, pick row, or tray moved

- **Visible cue:** the tote, a paper tray, or a glass still in the pick row is pushed, dragged, lifted, or turned during the episode; the tote is pulled toward the front edge to bring it within reach; or a gripper leans on the tote rim hard enough to shift it.
- **SOP rule broken:** Steps 1 to 5, the tote, the pick row, and the two paper trays stay where they are put.

- **Coaching note:** reach out to it. Never bring it to you.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a glass, pad, sheet, or the flag falls onto the desk or the floor short of its spot, a glass in the pick row is knocked over or struck, a standing pad is swept down, a tray is hit and shifts, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 5, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a glass still in the pick row or on the desk, a pair of glasses with no pad between them, a void unfilled, the flag off the ledge, an arm away from home, or an arm still over the tote.
- **SOP rule broken:** Step 6, confirm the packed tote and the empty pick row; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Glass already damaged** when the step begins: chipped, cracked, starred, or unable to stand on its own base. Replace it before the next episode.
- **Two glasses cannot be told apart by weight** from their shape, so the next glass in cannot be chosen. Discard the episode and restock the pick row with glasses whose weights are told apart at a glance.
- **Paper missing or short:** the base sheet is absent, the pad row runs out of interleave pads before the pick row is empty, or a paper tray runs out of void pads before every void is filled. Discard the episode and restock the paper.
- **Interleave pad will not stand** on its own foot on a flat surface even after it is refolded. Replace the pad before the next episode.
- **Tote does not fit the pack:** the glasses in the pick row will not go in in one row with a pad between every pair without touching. Discard the episode and restock the pick row or replace the tote.
- **Tote fault:** the tote is cracked, will not sit square, or its label ledge will not hold the flag flat. Replace it before the next episode.

- **Fragile flag missing, blank, or unreadable** when Step 5 begins.
- **Tote or glass slides** under a correct lower in because the desk is slick, wet, or crumbed.
Wipe and dry the desk before the next episode.

Annotation subtasks (from SOP)

1. Lay the base sheet on the tote floor
2. Move one glass from the pick row into the tote
3. Stand one interleave pad between two glasses
4. Fill one void with a void pad
5. Place the fragile flag on the tote ledge
6. Confirm the pack and end the episode